# Building a Mouse Lipidome Atlas with Cartridge-Based Nano-ESI and $\mu$PAC Separations

**Salma Abou Elhassan**[1], Li-Yu Chen[1], Katherine L. Walker[2], Graeme C. McAlister[2], Scott T. Quarmby[1], Katherine A. Overmyer[1,3], and Joshua J. Coon[1,3]

[1]University of Wisconsin-Madison, Madison, WI, USA; [2]Thermo Fisher Scientific, San Jose, CA, USA; [3]Morgridge Institute for Research, Madison, WI, USA

## Background

Insight into lipid regulatory networks requires sensitive, reproducible, and scalable MS data. Yet, discovery lipidomics is challenged by variability in ionization efficiency that impedes precise quantification. While nanoflow improves sensitivity and reduces sample consumption, it demands highly reproducible separations and stable spray.

Here, we use μPAC nanoflow chromatography to improve chromatographic resolution and retention time reproducibility by reducing eddy diffusion and band broadening. With μPAC chromatography and the Thermo Scientific™ OptiSpray™ Ion Source, we map a system-wide mouse tissue lipidome atlas, enabling reproducible lipid identification and quantification.

OptiSpray Ion Source

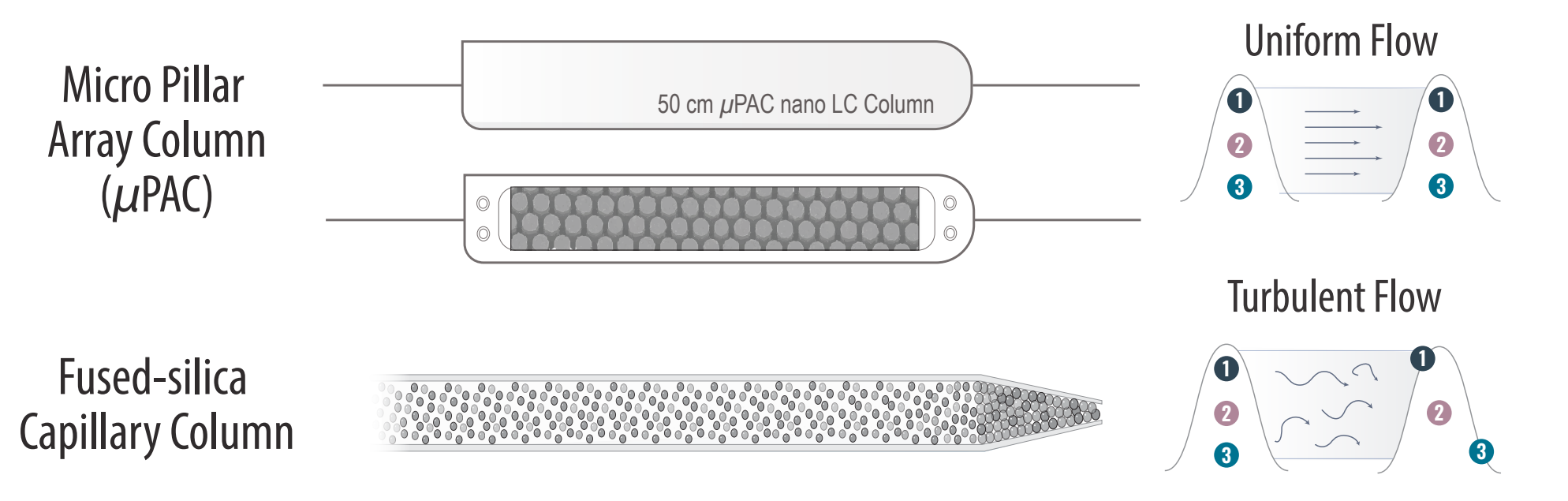

Micro Pillar Array Column (μPAC)
50 cm μPAC nano LC Column
Fused-silica Capillary Column
Uniform Flow
Turbulent Flow
1
2
3

Figure 1. Structural comparison between $\mu$PAC and capillary column

## Methods and Experiments

### Methods :

Separation was performed on a Vanquish Neo LC system using an OptiSpray $\mu$PAC Neo 50 cm cartridge mounted on an OptiSpray Ion Source and coupled to a modified Orbitrap Tribrid MS (Thermo Fisher Scientific). Mobile phases consisted of 0.2% formic acid in water **(A)**, and 0.2% formic acid, 5mM ammonium formate in (9:1) isoprpropanol /acetonitrile **(B)**.

Lipid separations were achieved using a 30-minute active gradient (60-min long run), acquired in separate positive and negative ionization runs. MS2 and MSn data were collected in data-dependent acquisition (DDA) mode with real-time library searching (RTLS) in the linear Ion Trap.

### Experiments :

1. Optimization of $\mu$PAC-OptiSpray Methods for nano-Lipidomics
2. Evaluating Methods' Robustness and Reproducibity
3. Building a System-Wide Mouse Tissue Lipidome Atlas

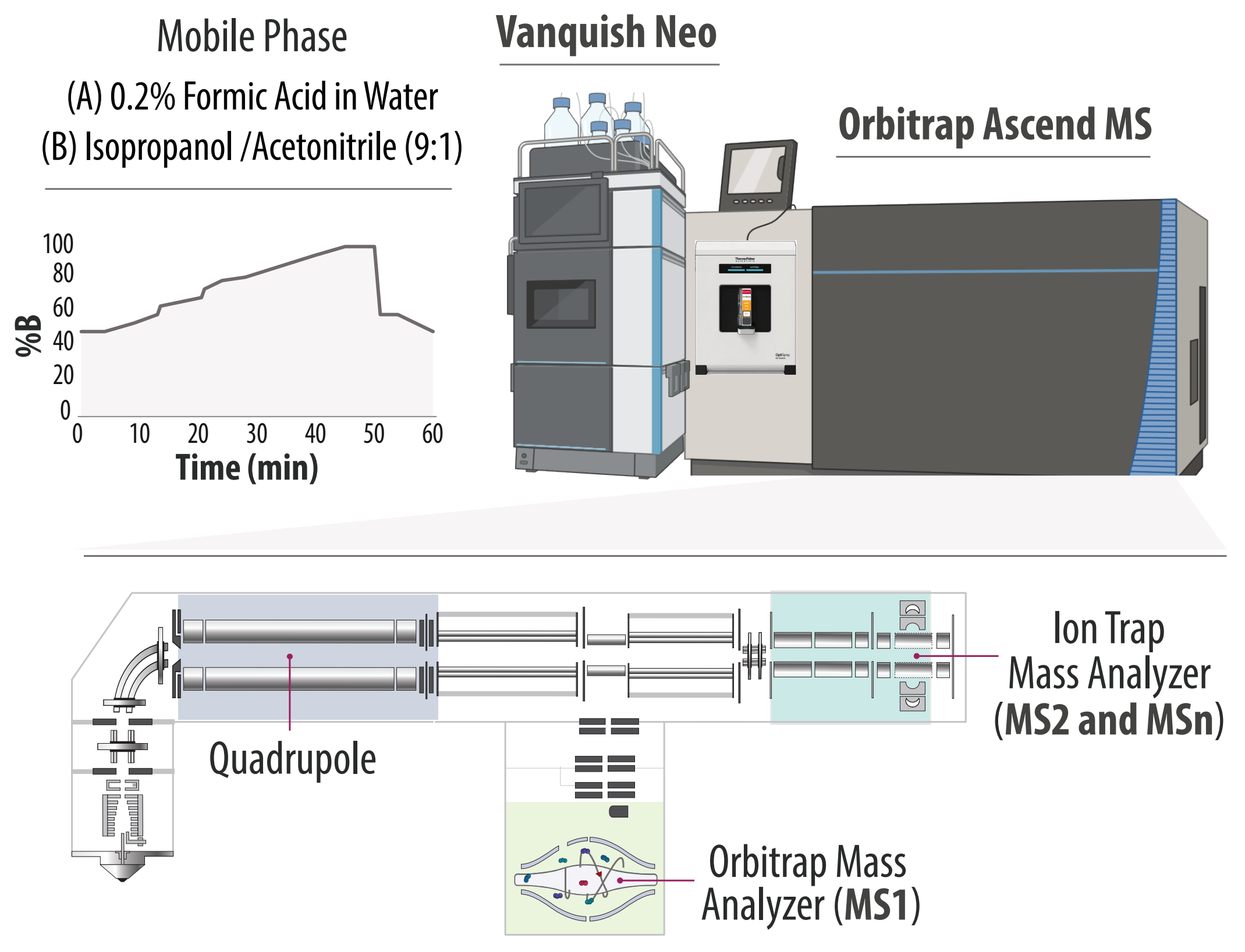
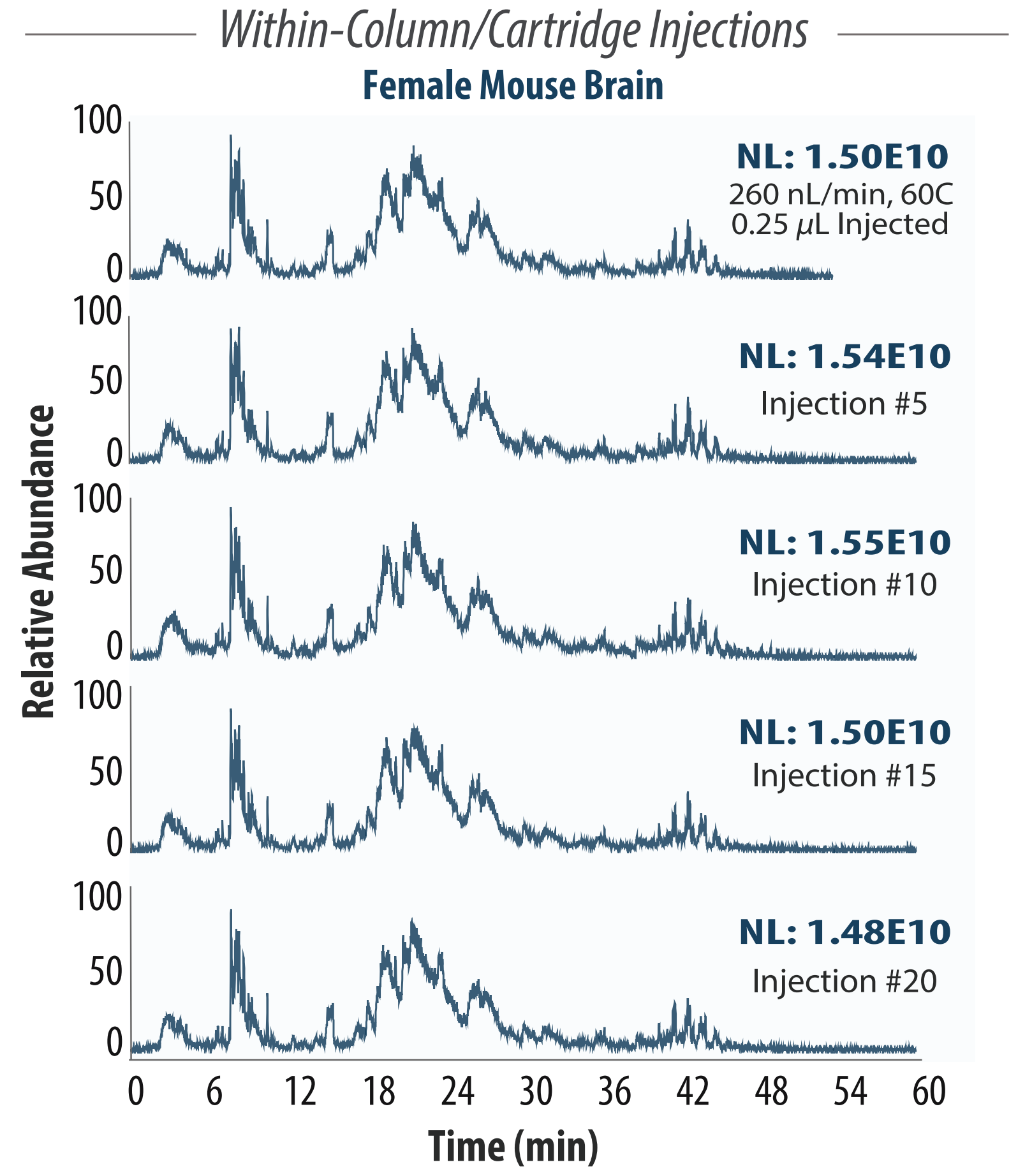

Mobile Phase
(A) 0.2% Formic Acid in Water
(B) Isopropanol /Acetonitrile (9:1)
%B
100
80
60
40
20
0
0 10 20 30 40 50 60
Time (min)
Vanquish Neo
Orbitrap Ascend MS
Ion Trap Mass Analyzer (MS2 and MSn)
Quadrupole
Orbitrap Mass Analyzer (MS1)

Figure 2. Schematic illustration of the Orbitrap Ascend Tribrid MS

## Column Reproducibility



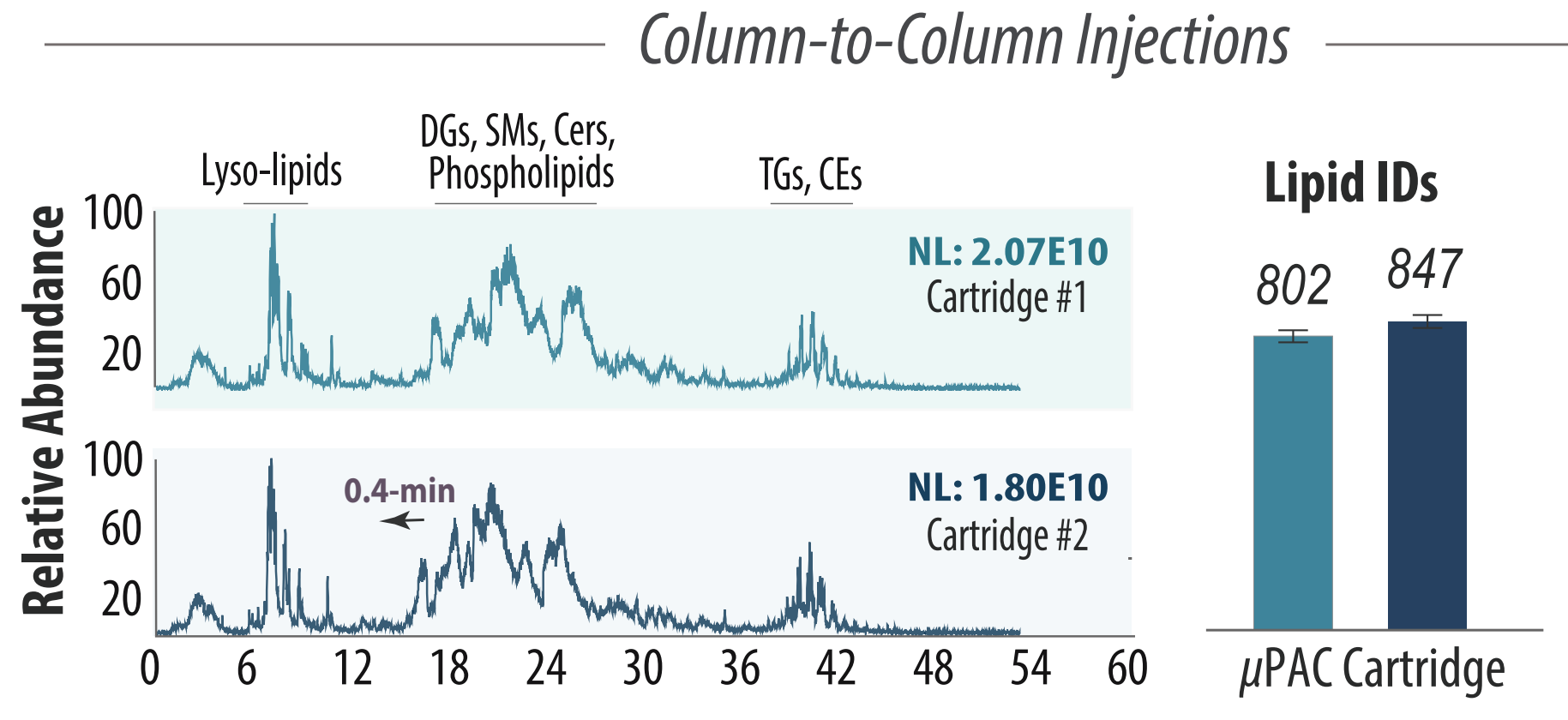


**Figure 4. $\mu$PAC column-to-column reproducibility**

Consistent lipid profiles and lipid IDs across $\mu$PAC cartridges.

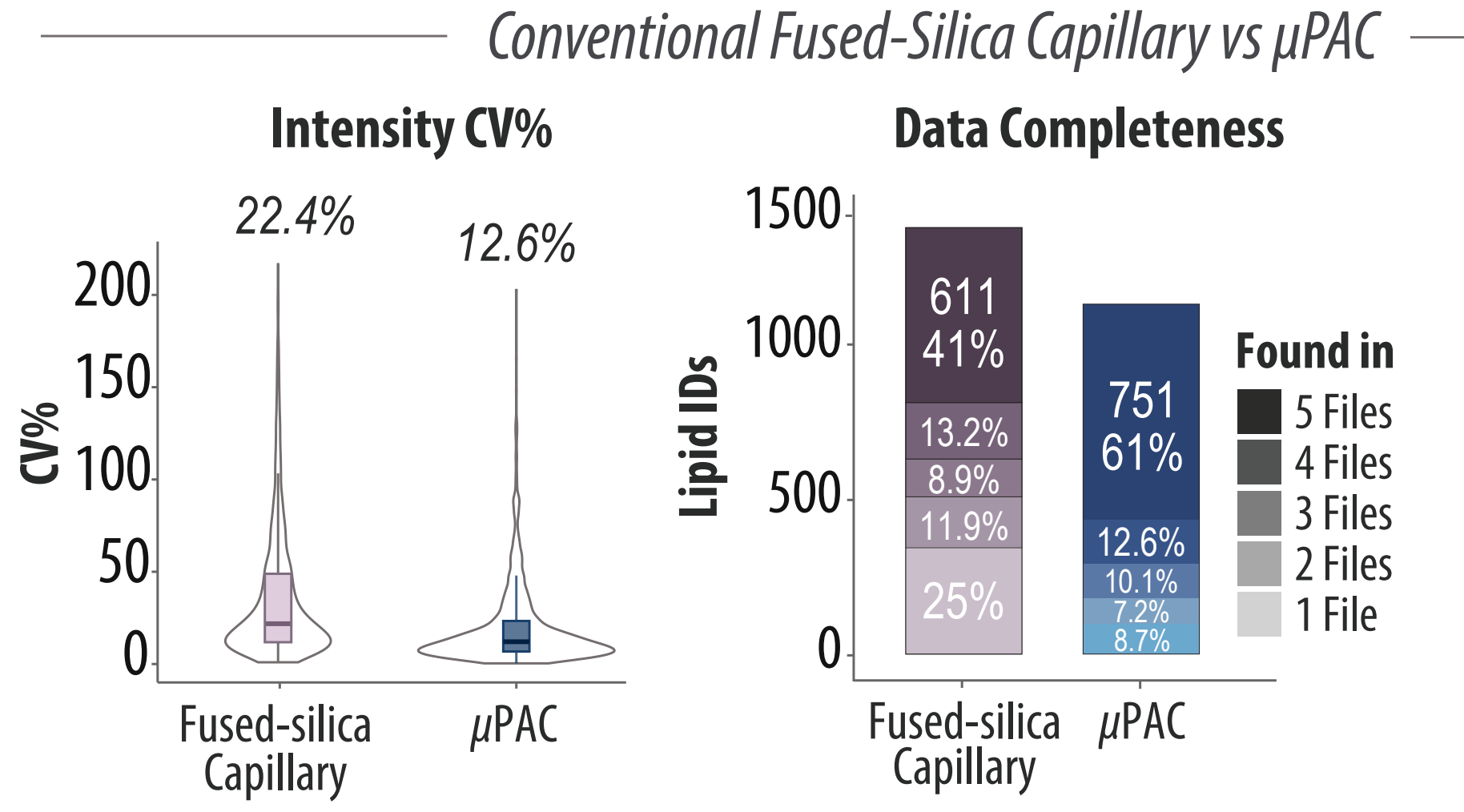


**Figure 5. Comparison between $\mu$PAC and fused-silica capillary columns**

$\mu$PAC showed lower intensity variability and highly reproducible data compared to conventional capillary columns.

## Conclusions

$\mu$PAC–OptiSpray nanoflow lipidomics improved lipid coverage and reproducibility compared to conventional capillary columns. The optimized workflow enabled reproducible construction of a mouse tissue lipidome atlas, revealing tissue- and sex-specific lipid differences across major biological systems.

### Future Directions

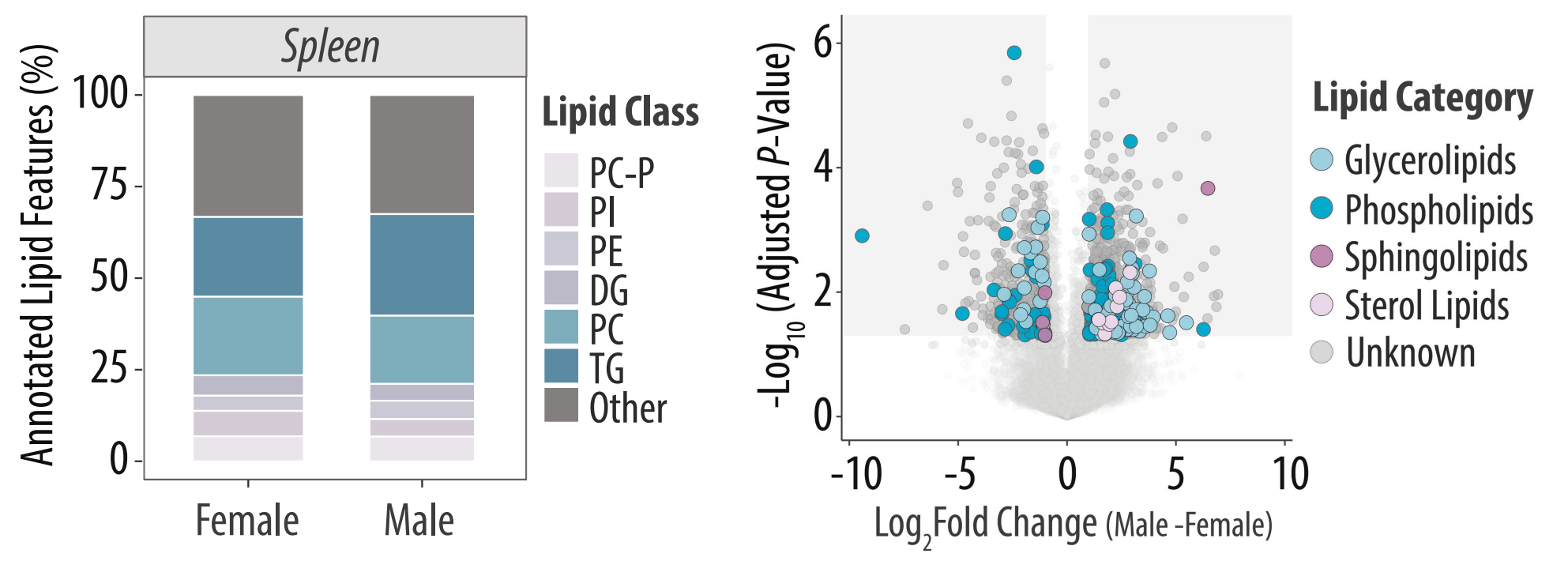
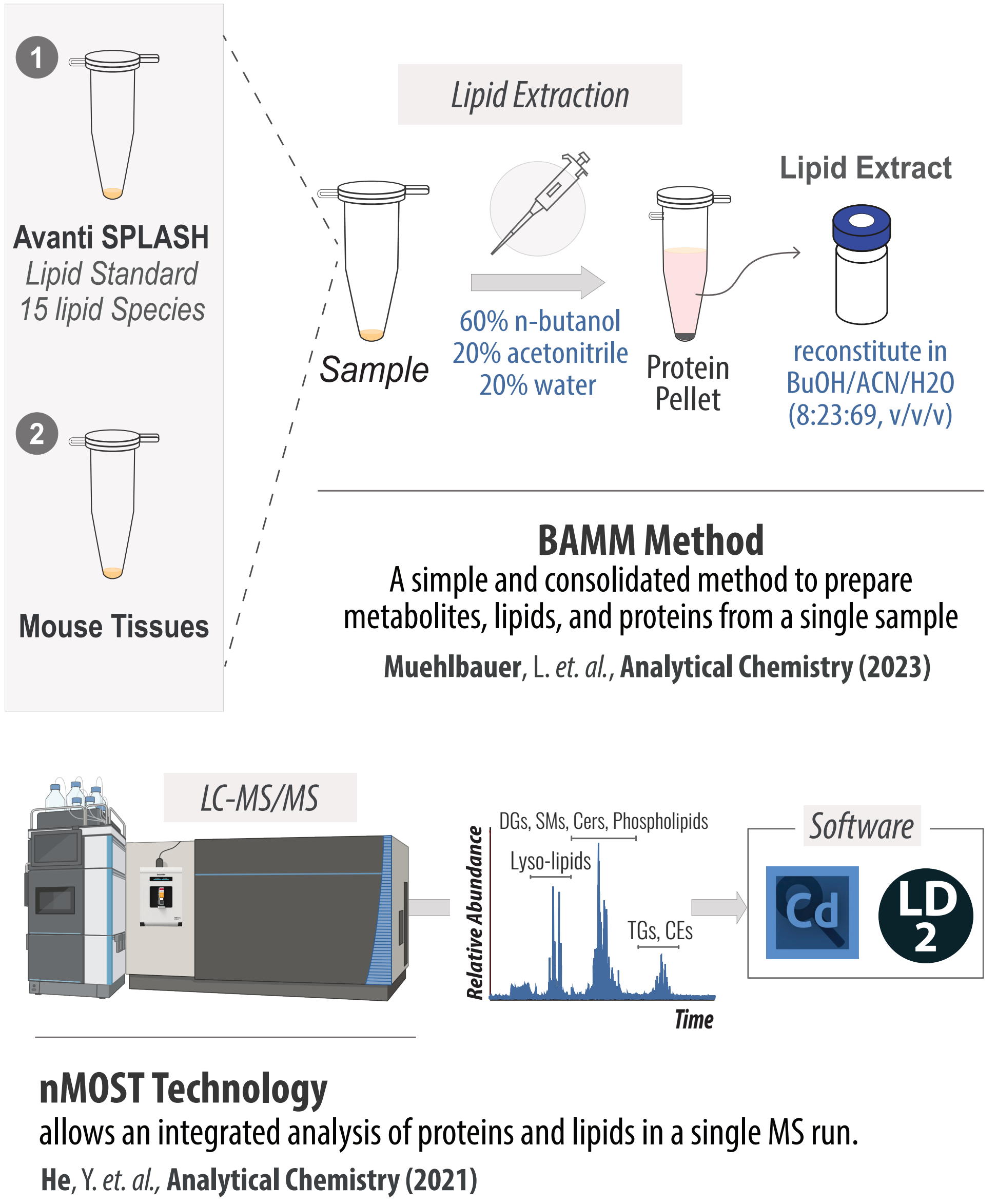


Figure 7. Sex-specific lipid composition and abundance differences in mouse spleen

## Workflow



**Avanti SPLASH**
*Lipid Standard*
*15 lipid Species*

**Mouse Tissues**

### BAMM Method

A simple and consolidated method to prepare metabolites, lipids, and proteins from a single sample

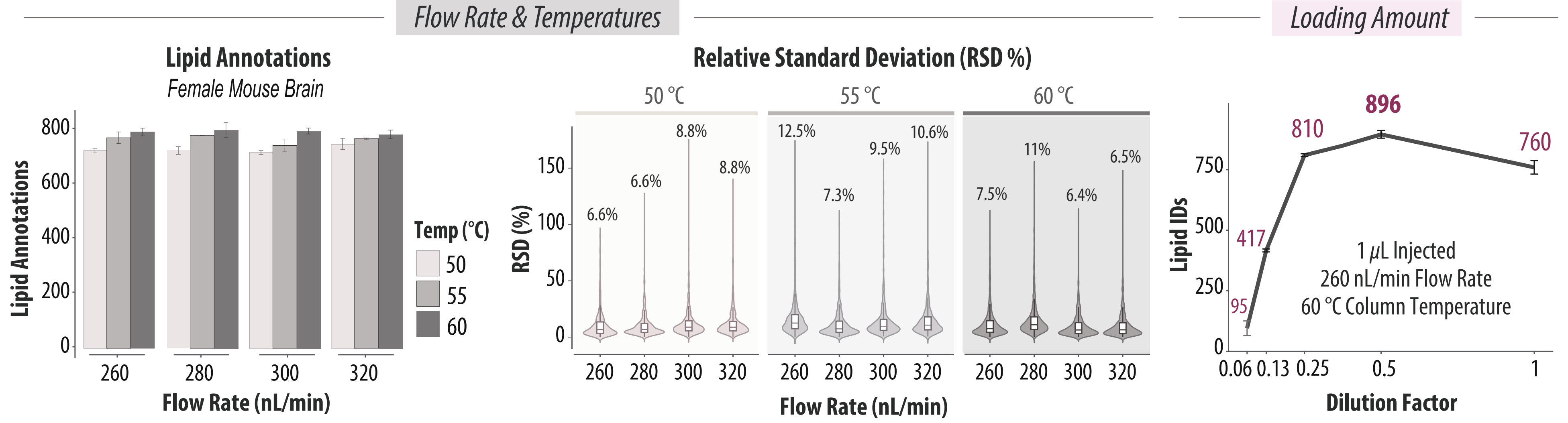
**Muehlbauer**, L. *et. al.*, **Analytical Chemistry (2023)**

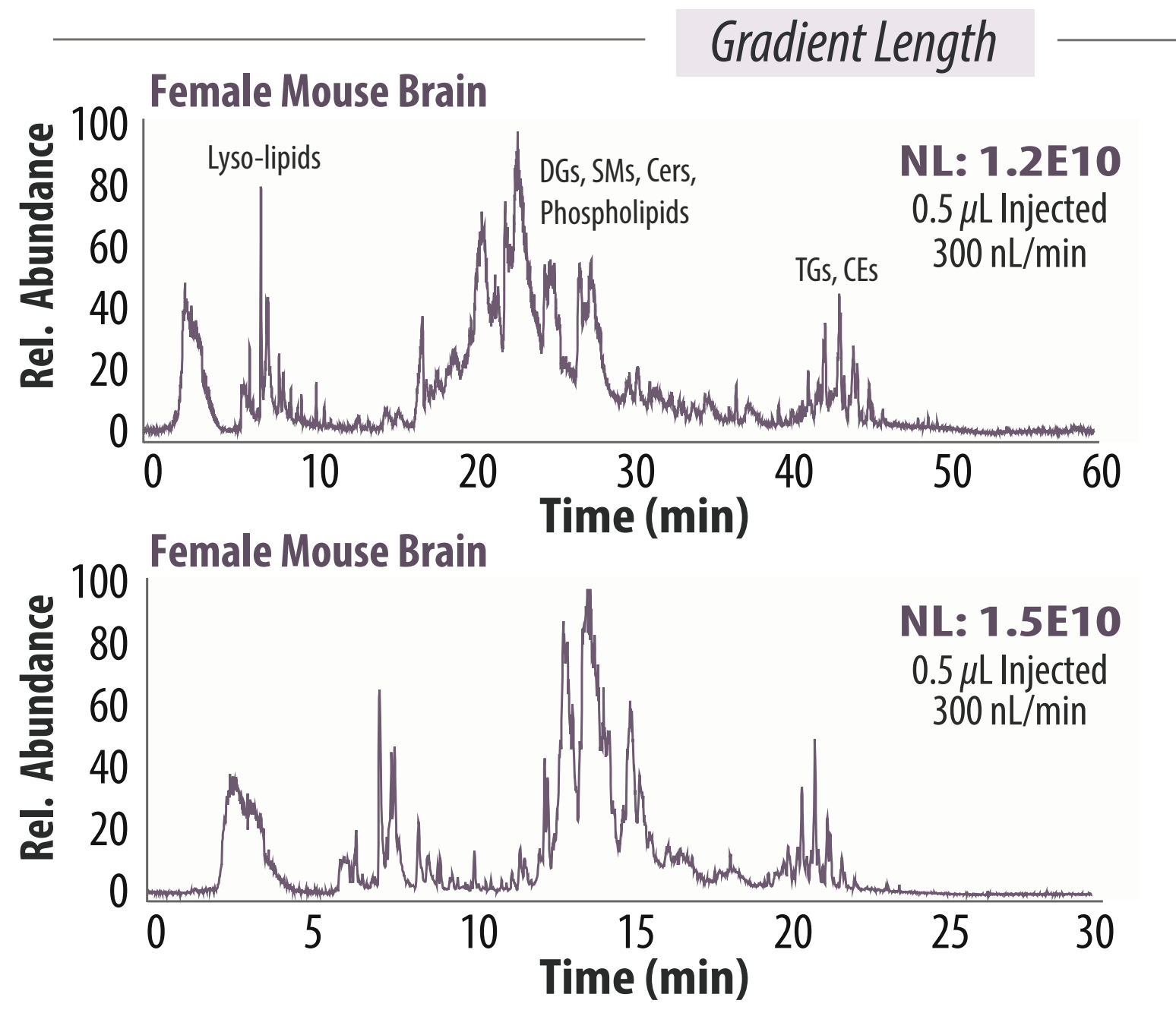
### nMOST Technology

allows an integrated analysis of proteins and lipids in a single MS run.

**He**, Y. *et. al.*, **Analytical Chemistry (2021)**

## Method Optimization





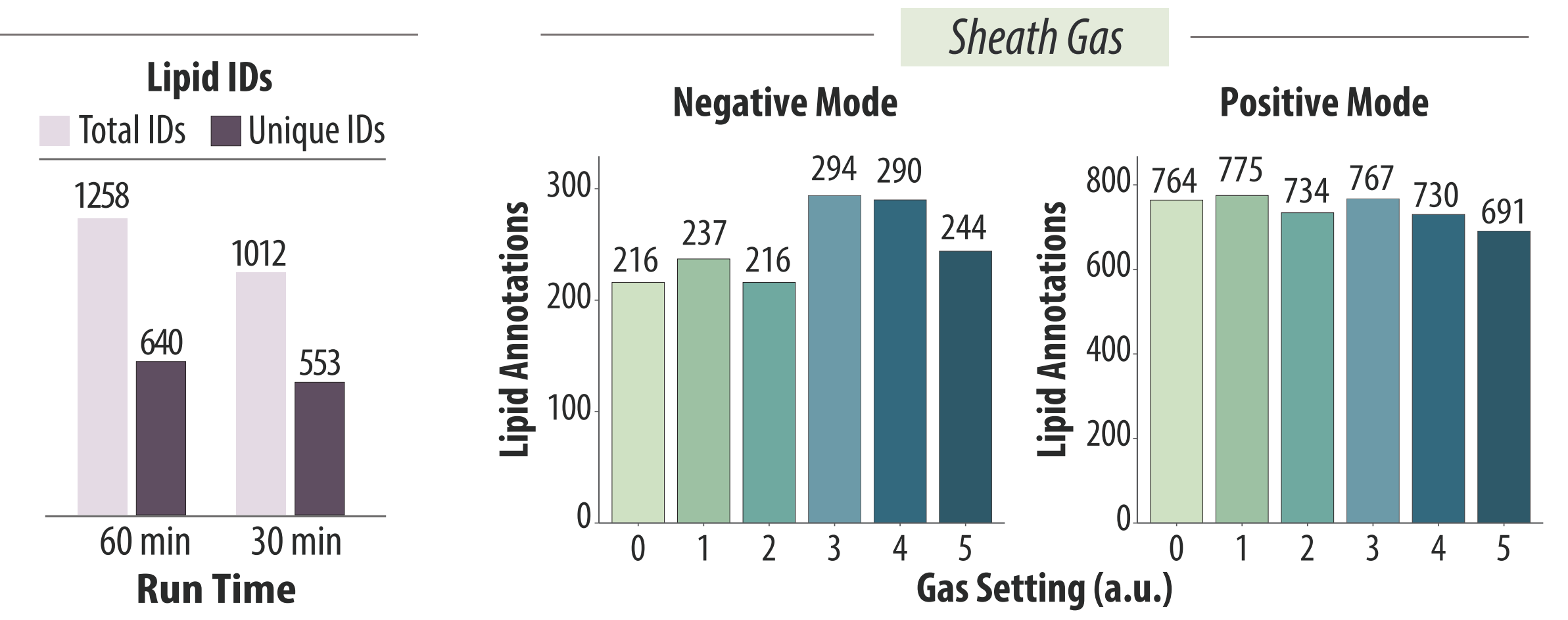
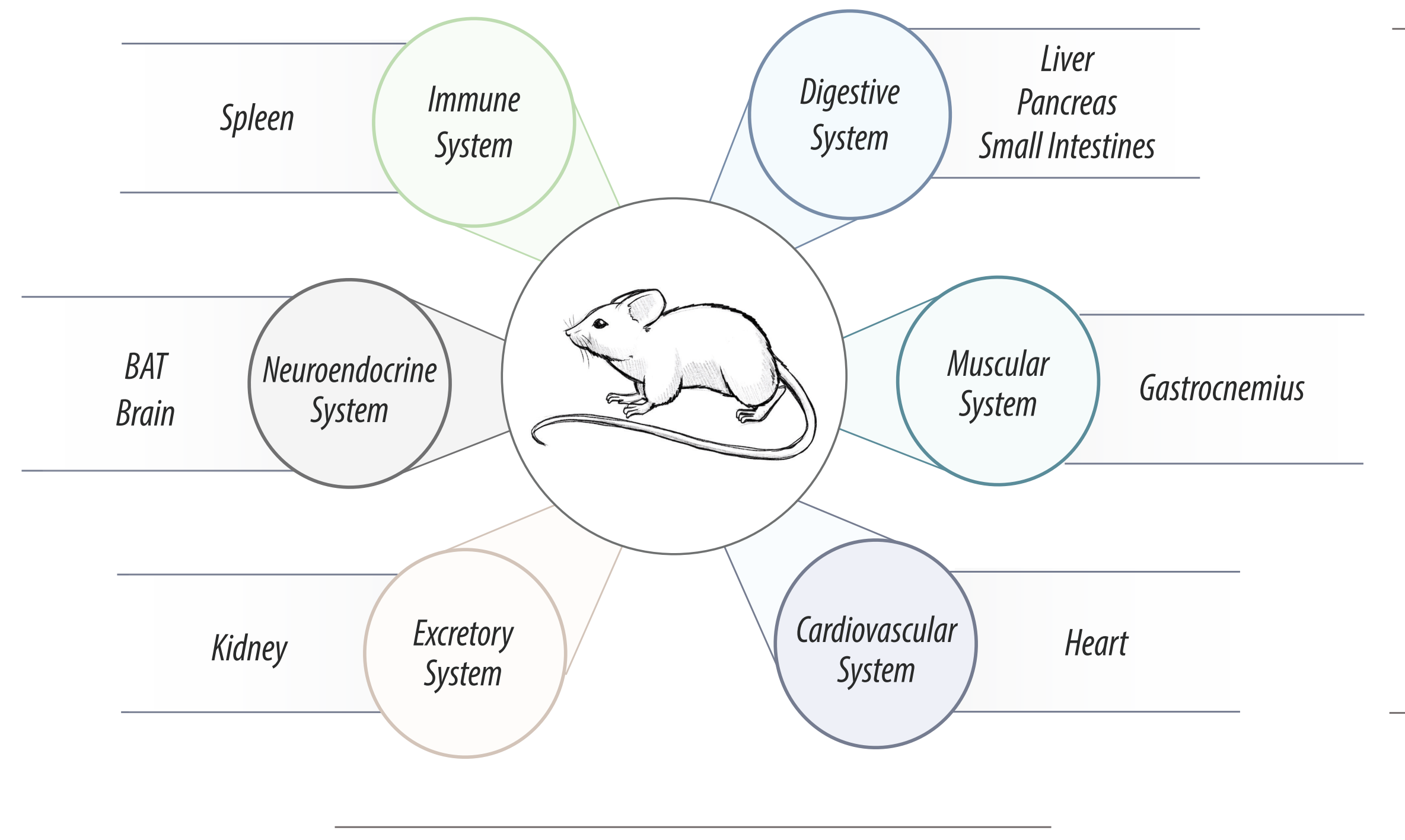


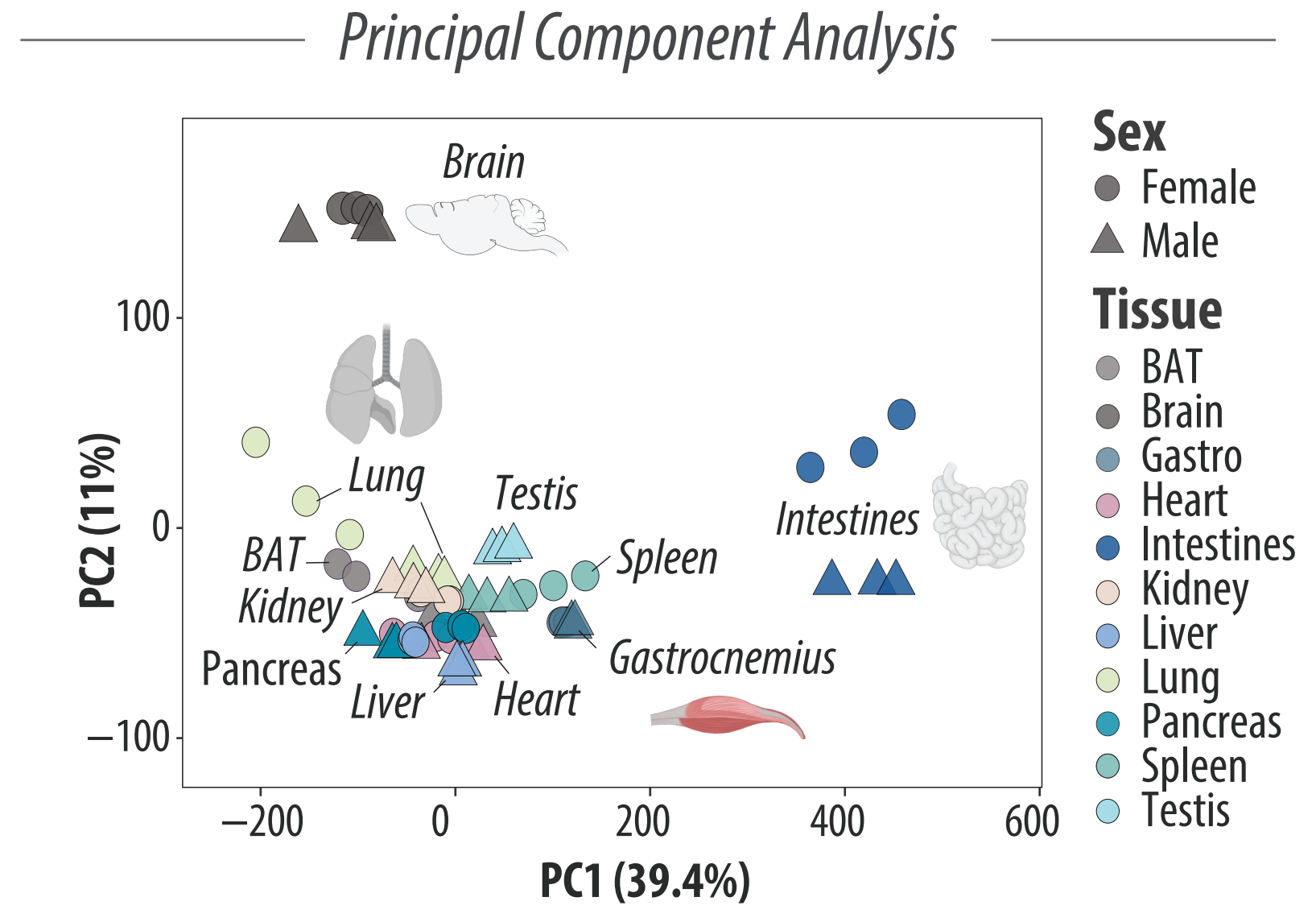
**Figure 3. Method optimization for $\mu$PAC–OptiSpray lipidomics**

Lipid annotations were compared across flow rate, column temperature, loading amount, gradient length, and sheath gas settings. A column temperature of 60 °C gave the highest lipid annotations, sheath gas improved stability in negative ion mode, and the shorter 30 min gradient retained ~87% of unique lipid IDs while reducing run time.

## Lipidome Atlas of Mouse Tissues



**Mouse Lipidome Atlas**



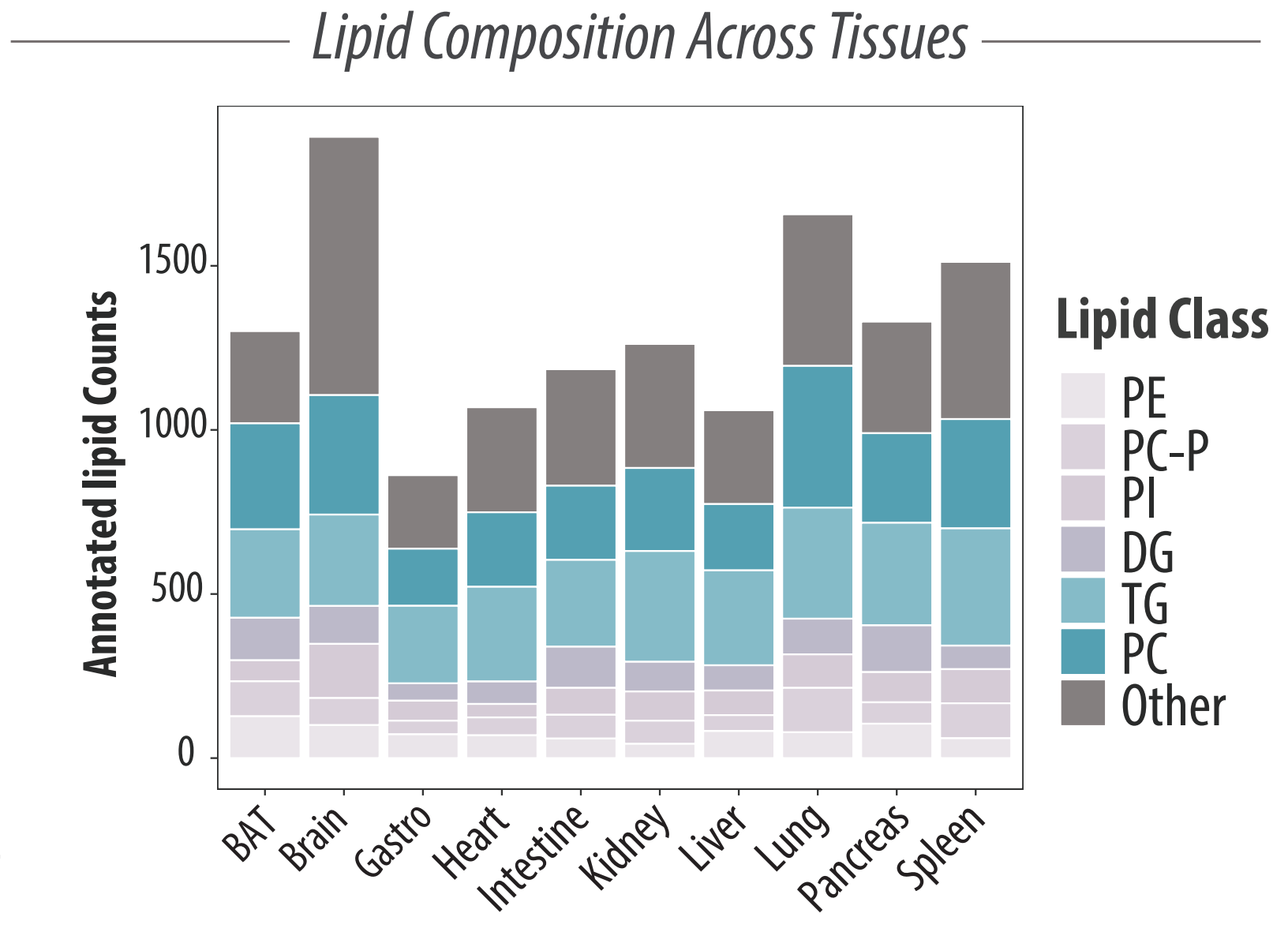


**Figure 6. Mouse lipidome atlas using $\mu$PAC–OptiSpray nanoflow lipidomics**

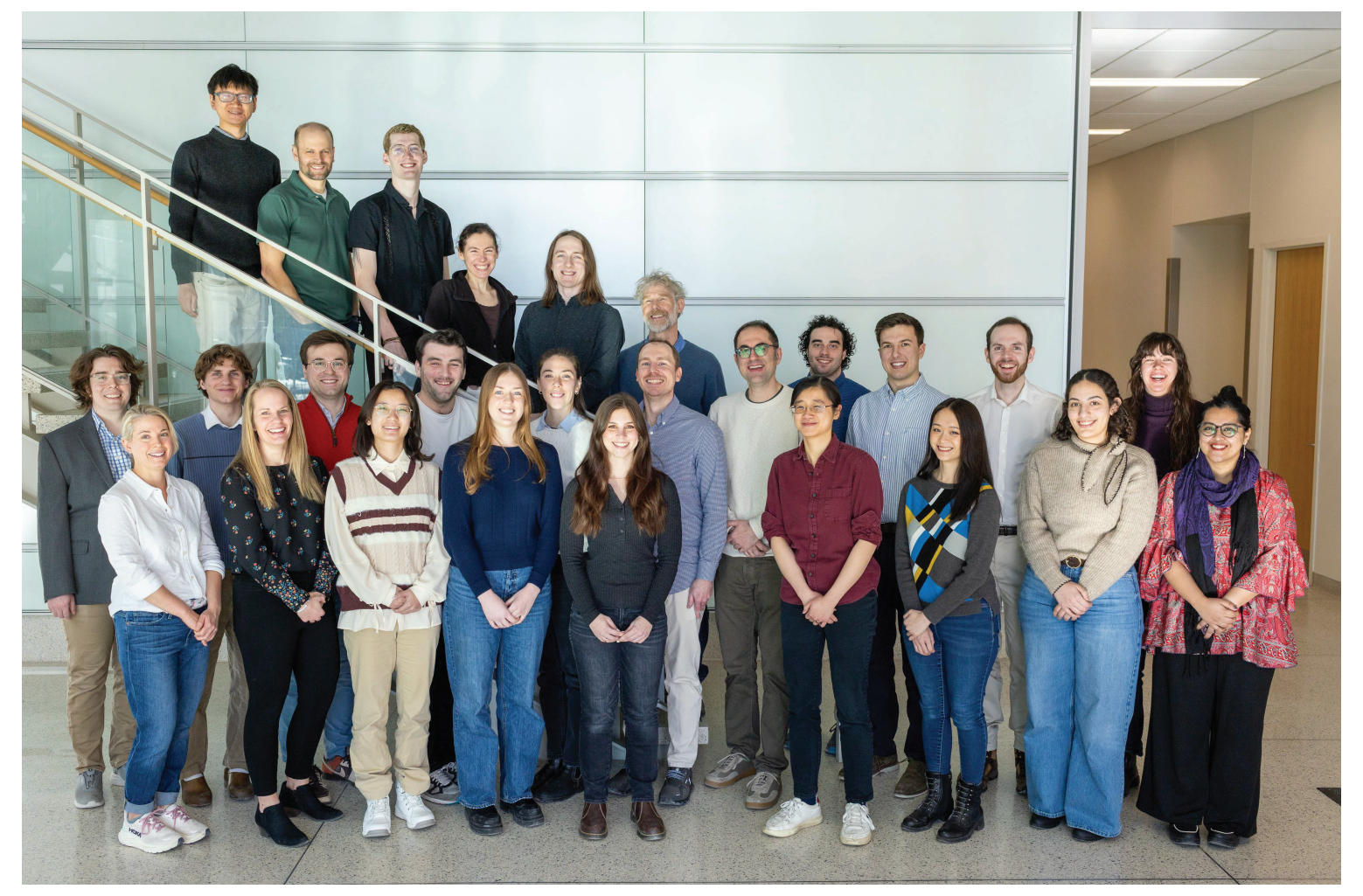
OptiSpray-based lipidomics workflow was used to build a reproducible mouse tissue lipidome atlas from male and female mice. Tissues representing immune, digestive, muscular, cardiovascular, excretory, and neuroendocrine systems showed clear separation by PCA, indicating distinct tissue-specific lipid profiles.

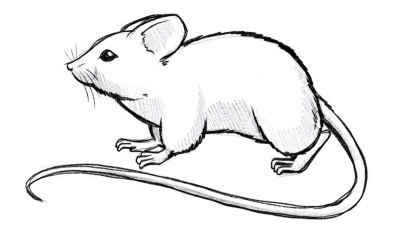
Lipid class distributions varied across tissues, with brain and lung showing high lipid annotation counts, intestines enriched in TG-related signal, and tissues such as brain, lung, and spleen showing prominent PC contributions. These results highlight broad lipidome diversity across mouse tissues and support OptiSpray as a reproducible platform for comparative tissue lipidomics.

## Acknowledgements

We thank the National Institutes of Health and Thermo Fisher Scientific for support. Work in this project was supported by the National Science Foundation under grants NIH/NIA P41 GM108538.

**Conflict Statements:**
JJC is a consultant for Thermo Fisher Scientific and Seer. KLW and GCM work for Thermo Fisher Scientific who make and sell the sources, columns and mass spectrometers used in this study. Also, this study utilized some material support from Thermo Fisher Scientific via a combination of collaborative research and material transfer agreements.





## References

- **Muehlbauer**, Laura K., *et al.* "Rapid multi-omics sample preparation for mass spectrometry." **Analytical Chemistry** 95.2(**2023**): 659-667.
- **He**, Y, *et al.* "Multi-Omic Single-Shot Technology for Integrated Proteome and Lipidome Analysis." **Analytical Chemistry** 93.9 (**2021**) 4217-4222.
- **Anderson**, Benton J., *et al.* "LipiDex 2 Integrates MSn Tree-Based Fragmentation Methods and Quality Control Modules to Improve Discovery Lipidomics." **Analytical Chemistry** (**2024**).
- **Brademan**, Dain R., *et al.* "Improved Structural Characterization of Glycerophospholipids and Sphingomyelins with Real-Time Library Searching" **Analytical Chemistry** 95.2 (**2023**):7813-7821